

Keon Systems

The Constitutional AI Execution Substrate

Version 2.0 · Status: Canonical Architecture

Runtime Source of Truth: keon-systems/docs/governance/*.md · keon-systems/launch/keon-v-launch-manifest.yaml

1 The Problem: Trust-Me Architecture

Traditional AI agent frameworks execute before authorization and rely on post-execution logging. If logging fails, the system cannot prove what occurred. Divergence between side effects and audit trails creates legal and operational liability.

Keon makes out-of-policy execution structurally impossible before it begins.

2 The Standard: Cryptographically Governed AI Execution (CGAE)

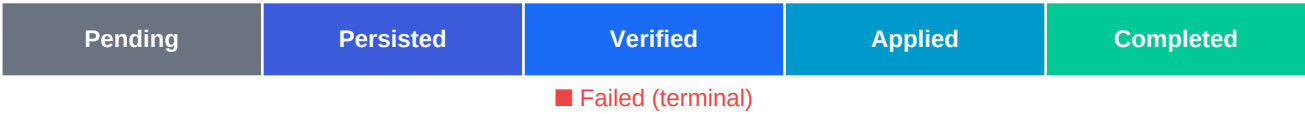
Architectural Invariants

I-EXEC-01	Execution SHALL NOT proceed unless authorization receipt is durably persisted prior to side effects.
I-MEM-01	Memory mutation SHALL NOT occur without a corresponding persisted memory receipt.
I-LEDGER-01	Ledger entries SHALL be append-only and immutable within a partition.
I-VERIFY-01	All governance artifacts MUST be verifiable offline, independent of runtime.

3 The Substrate: ARO, JCS, and Keon Memory

The **Authoritative Receipt Outbox (ARO)** enforces receipt-gated state mutation. All governed state transitions route through ARO and fail closed. Keon applies RFC 8785 JSON Canonicalization Scheme (JCS) before hashing. Receipts are signed using Ed25519 and are verifiable offline. **Keon Memory** is an append-only, UUIDv7-ordered, partition-scoped constitutional ledger. Receipts, traces, and evidence are ledger-derived.

ARO State Machine



4 ALPHA Authority Doctrine

Letter	Meaning	Description
A	Attested	Cryptographically signed, canonicalized, non-repudiable
L	Ledger	Append-only causal spine (Keon Memory)
P	Policy-Bound	Tied to deterministic PolicyHash
H	Human	Explicit human authorization domains
A	Authority	Binding governance transitions

No authority. No execution. No exceptions.

5 Governed Execution (GE-001)

Keon exposes governance through a suite of MCP tools: `keon.governed.execute`, `keon.runs.get`, `keon.runs.stream`, `keon.evidence.export`, and `keon.evidence.verify`.

Execution Lifecycle



6 Evidence Packs

A completed governed run produces an **Evidence Pack** containing signed receipts, policy context, and Spine references. Evidence Packs are verifiable offline using `keon-cli`.

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Tamper detection produces non-zero exit codes

If the Evidence Pack verifies — the execution was governed.

If it does not verify — it was not.

There is no third state.

7 Constitutional Trust Model

Keon does not require trust in its runtime. It requires only verification of output. Receipts are cryptographically bound, ledger-derived, and independently auditable.